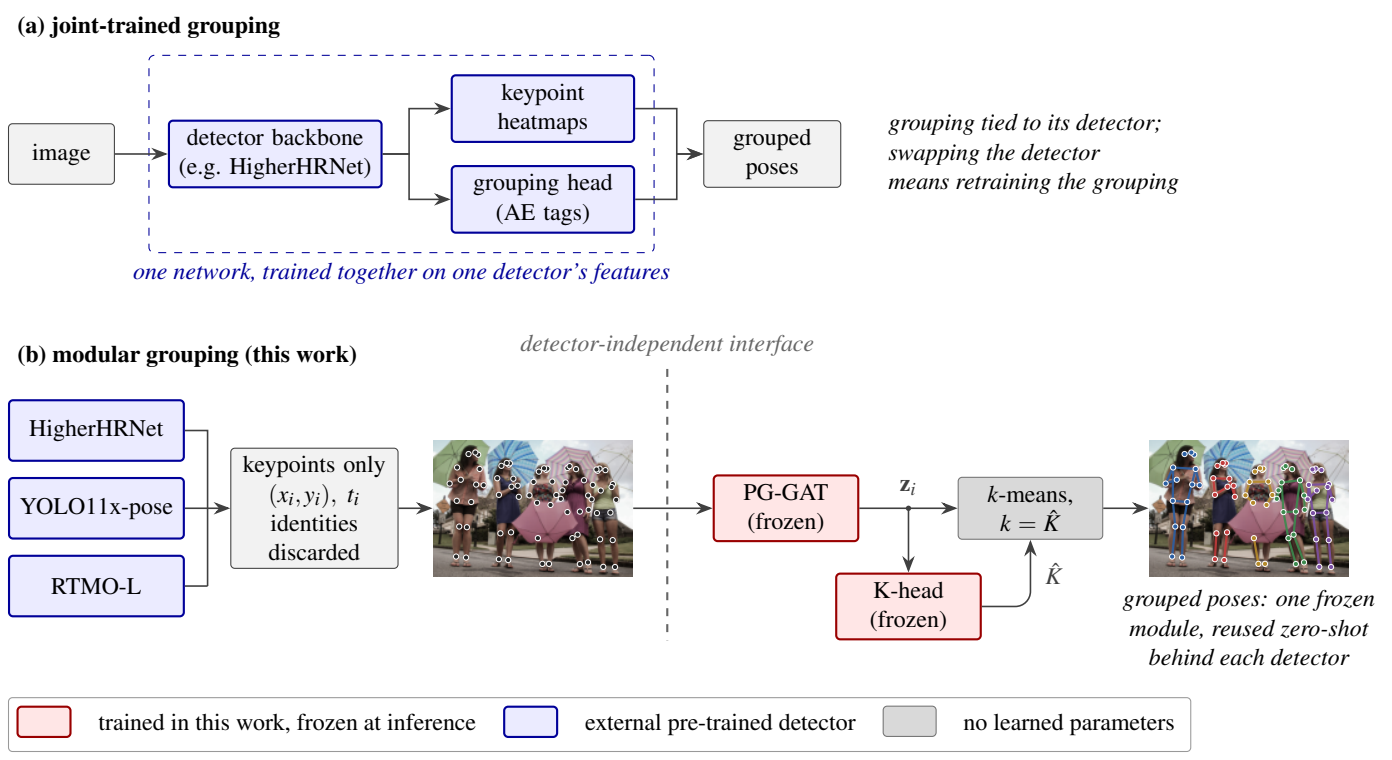

(a) joint-trained grouping
image
detector backbone (e.g. HigherHRNet)
keypoint heatmaps
grouping head (AE tags)
grouped poses
grouping tied to its detector; swapping the detector means retraining the grouping
one network, trained together on one detector's features
(b) modular grouping (this work)
detector-independent interface
HigherHRNet
YOLO11x-pose
RTMO-L
keypoints only (x_i, y_i), t_i identities discarded
PG-GAT (frozen)
$\mathbf{z}_i$
k-means, $k = \hat{K}$
K-head (frozen)
$\hat{K}$
grouped poses: one frozen module, reused zero-shot behind each detector
trained in this work, frozen at inference
external pre-trained detector
no learned parameters